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Advanced Apex Project II

Week 1: Project Orientation & Problem
Statement Selection

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What is Advanced Apex Project II?

Advanced Apex Project II is an experimentation-oriented Machine Learning project where students:

- Work on real-world datasets
 - Apply machine learning techniques
 - Compare multiple approaches
 - Perform experimentation and tuning
 - Interpret results
 - Document findings professionally
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Apex-2 End-to-End Workflow

Problem Statement Selection



Dataset Selection



Data Understanding



Data Cleaning & Preprocessing



Exploratory Data Analysis (EDA)



Feature Engineering



Model Building



Hyperparameter Tuning



Interpretation & Documentation



Final Submission

Types of Machine Learning Projects

Project Type	Goal	Output
Classification	Predict category	Yes/No, Class Label
Regression	Predict value	Numeric Value
Clustering	Discover groups	Segments
ANN	Neural-network based prediction	Classification / Regression

Classification Projects

Objective

Predict a category or class label

Common Algorithms

Logistic Regression

Decision Tree

Random Forest

Gradient Boosting

Examples

Employee leaves or stays?

Customer accepts offer or not?

Product is defective or not?

Output Example

Attrition = Yes

Attrition = No

Regression Projects

Objective

Predict continuous numerical values

Common Algorithms

Linear Regression

Random Forest Regressor

Gradient Boosting Regressor

Examples

Sales amount prediction

House price prediction

Revenue forecasting

Output Example

Predicted Sales = ₹12,750

Clustering Projects

Objective

Discover hidden groups within data

Common Algorithms

K-Means

Hierarchical Clustering

DBSCAN

Examples

Customer Segmentation

Market Segmentation

User Behaviour Analysis

Output Example

Cluster 1 → Premium Customers

Cluster 2 → Occasional Buyers

ANN Projects

Artificial Neural Networks

Inspired by biological neurons.

Key Components

Input Layer

Hidden Layers

Output Layer

Activation Functions

Backpropagation

Applications

Customer Response Prediction

Product Quality Classification

Advantages

Learns complex patterns automatically

Expected Deliverables

Students are expected to perform:

- Data Selection
- Data Preprocessing
- Exploratory Data Analysis
- Feature Engineering
- Model Building
- Hyper parameter Tuning
- Model Evaluation
- Interpretation
- Comparison of Techniques
- Documentation

Why Experimentation Matters

Good Practice

1. Train multiple models:
 - Logistic Regression
 - Decision Tree
 - Random Forest
2. Compare performance.
3. Tune parameters.
4. Document observations.

Goal

Find the most appropriate solution rather than the first solution.

Importance of Interpretation

A model result alone is not enough.

Students should explain:

What happened?

Why it happened?

Which features influenced predictions?

What business insights can be derived?

Example

Employee satisfaction and overtime significantly influence attrition risk.

Dataset Sources

Kaggle

Benefits:

- Thousands of datasets
- Community notebooks
- Industry-oriented datasets

UCI Machine Learning Repository

Benefits:

- Academic benchmark datasets
- Clean datasets
- Widely used in research

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Thank You
